

PRODUCT REQUIREMENT DOCUMENT

Markz Voice: Hands-Free Classroom Ledger v2.0

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Target Audience:	Karnataka State Syllabus School Teachers

1. Product Overview & Core Vision

Markz Voice (v2.0) is a mobile-first, offline-first digital ledger tailored directly for teachers working within the Karnataka State Board syllabus ecosystem. Traditional post-test processing requires manual compilation, arithmetic summation, percentage generation, and alphabet-grade mapping for large class blocks (60-70 students per classroom).

Markz Voice replaces physical calculator workflows and micro-touch typing apps with a completely hands-free Voice Intake engine. By keeping their eyes and hands directly on the student physical answer scripts, teachers dictate sequential score matrices smoothly, eliminating high-frequency cognitive context switching and accelerating ledger submission timelines safely.

2. Target Persona & Problem Definition

User Persona

The typical user is a high school teacher in a tier-2 or tier-3 region in Karnataka managing large classroom contexts (60+ student rosters). While fully fluent in common media applications (such as WhatsApp or YouTube), they express deep friction towards multi-input data-entry configurations on mobile fields.

Core Problem Workflow Context

A typical evaluation timeline contains the following structural steps:

1. Evaluate physical script paper bundle sequentially.
2. Jot marks manually on the paper face sheet per subject element.
3. Open physical ledger books; match names or roll ids.
4. Fire up a physical or phone desk-calculator to sum marks across 4-6 foundational streams.
5. Convert aggregate totals to specific percentages manually via standard operational cross-multiplication math.
6. Reference KSEEB/state criteria to map explicit letters (A+, A, B+, etc.) to individual scores and final metrics alike.
7. Transcribe metadata structures into official paperwork logs manually.

This repetitive process is highly prone to data entry transcription errors, and single typographical changes risk ruining the mathematical balance of an entire spreadsheet row matrix.

3. Core Technical Constraints

- **Local, Device-Bound NLP:** Data processing, voice tokenization, and numerical recognition parsing must operate entirely offline on the client hardware instance. Reliance on persistent cloud-side processing is functionally prohibited due to volatile connectivity constraints in rural or semi-urban staff rooms.
- **Local Accent & Dialect Adaptation:** The acoustic processing models must natively accommodate Indian English numbering styles and clear regional speech tones without crashing on missing accents (e.g., parsing variations like 'double two' vs 'twenty-two' or handling localized numeric cadence flawlessly).
- **High Acoustic Isolation thresholds:** Classroom staff rooms display significant background noise floors. The app requires localized wave-filtering optimizations and should encourage standard wired audio headset peripherals during peak dictation pipelines.

4. Functional Requirements & Product Epics

Epic 1: Test Template Configuration (Tap-Based Init)

Before initiating high-speed voice recording streams, the operator explicitly handles configurations through quick structural setup steps once:

- **Configuration Parameters:** Select total subject count metrics (e.g., 4 to 6 core subjects) and establish absolute Maximum Marks values (e.g., 25, 50, or 100) per block element.
- **Execution Sequence Mapping:** Define a locked sequential parsing matrix order (e.g., English -> Kannada -> Mathematics -> General Science) so the internal tracking logic knows exactly how spoken tokens line up automatically.

Epic 2: Continuous Voice Intake Engine

- **Uninterrupted Voice Mode:** Teachers trigger a specialized multi-row speech session. The system captures input across consecutive records without requiring physical touch interactions between subjects.
- **Dynamic Speech Token Parsing:** Teachers utter natural conversational rows (e.g., 'Roll 1: twenty-one, twenty-three, twenty-four, twenty-one'). The parsing pipeline maps individual slots dynamically to active subject indices based on native structured pauses.
- **Instant Mathematical Computation:** Upon registering the final defined subject marker, the math system evaluates equations immediately without backend delays:

$$\text{Subject Percentage} = (\text{Spoken Mark} / \text{Max Marks}) \times 100$$

Grand Total Marks = Σ (Spoken Marks)

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Overall % = (Total Marks / Σ Max Marks) × 100

• Real-time Audio Status Callback: An offline Text-to-Speech (TTS) voice confirms updates instantly so operators never look at screen fields. (e.g., Device speaks clearly: 'Total 89, Grade A. Roll 2 next').

Epic 3: Voice-Based Error Correction & Boundary Safeguards

- Interactive Deletion Hooks: Spoken macro triggers such as 'Clear Row' or 'Correction' scrub the current row state instantly, moving tracking focuses cleanly to Subject 1 of the active index.
- Hard Boundary Validations: If spoken marks surpass set criteria (e.g., uttering 'twenty-seven' against a max margin cap of twenty-five), the system soft-halts, fires an audio alert chime, and states 'Invalid marks detected for Mathematics, please repeat execution'.

5. Voice Interaction Architecture Matrix

The detailed transactional interface follows this deterministic interaction model:

Teacher Dictates (Input Phrase)	System Engine Processing State	Text-to-Speech (TTS) Feedback String
"Roll 1"	Instantiates Row 1 framework; binds tracking point to Subject 1 cell tracking block.	*Chime* / "Roll 1"
"21, 23, 24, 21"	Parses array items. Validates numbers, runs core calculations, saves metrics locally.	[Processing...] "Total 89. Grade A. Roll 2?"
"22, 15, 12, Absent"	Intercepts 'Absent' string. Flags raw matrix cell index as AB (0 marks evaluated).	"Total 49. Grade C. Roll 3?"
"Correction" or "Clear"	Flushes buffer data fields for current row indices cleanly.	"Roll 3 cleared. Ready for input."

6. UI/UX Dashboard Principles

- Maximized Structural Layout Density: Display high-contrast text fields explicitly presenting the current listening state and tracking targets prominently so checking data out of the corner of an eye is effortless.
- Real-time Waveform State Metrics: Provide clean, non-obtrusive, highly fluid visual cues that provide immediate confirmation that audio capture mechanics are running soundly.
- Color-Coded Performance Audits: Group final outputs dynamically using recognizable status

palettes (e.g., Soft Greens for elite milestones, Muted Yellows/Oranges for mid-tier marks, and Alert Reds highlighting critical remedial indicators).

7. Out of Scope Boundaries (V1 Release Explicitly)

- Fluid Context Conversational Parsing: Processing unstructured statements like 'Hey look, put twenty points in Kannada and for English give him ten' is deferred. Input formatting requires structured parsing cadence sequences to lock down 99.9% processing accuracies.
- Multi-Device Cloud Synchronization Architecture: Multi-node replication models, cross-device authorization flows, and remote cloud infrastructure architectures are pushed entirely to secondary iterations. All operational files compile natively on local sqlite blocks.